

Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from Overseas to the East Coast of India

Problem Statement Source	Smart India Hackathon (SIH) — Logistics & Transportation
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1. Problem Statement

Bulk cargo procurement for India's East Coast ports — including Paradip, Visakhapatnam, Gangavaram, Gopalpur, Dhamra, Sagar-Sandheads, and Haldia — currently relies on daily, reactive engagement with the freight market rather than a forward-looking, data-driven strategy. Charterers evaluate the spot market on a day-to-day basis instead of forecasting rate movements, which results in three recurring problems.

- **No predictive visibility into freight rates**, making it difficult to time short-term or mid-term charter contracts to favourable market windows for routes from Australia, the United States, Mozambique, Russia, and Indonesia.
- **Suboptimal vessel-type selection** (Handysize, Supramax, Panamax, or Capesize) for a given cargo parcel, since draft, LOA, beam, and cargo-handling constraints at both the loading port and the destination East Coast port are not systematically cross-checked against forecasted market conditions.
- **Avoidable vessel idle time and deadheading**, caused by the absence of a mechanism that anticipates periods of low demand or port congestion and proposes alternative positioning before the fact.

The compounding effect of these gaps is higher freight cost per tonne, inconsistent turnaround planning, and exposure to market and port-side shocks that could otherwise be forecast and hedged against. The objective is to move the chartering workflow away from repeated single-voyage spot fixtures and toward planned short-term/medium-term multi-voyage contracts, supported by quantitative forecasts rather than daily market sentiment alone.

2. Proposed Solution

The proposed system is a machine-learning-driven Freight Forecasting and Decision Support Platform that ingests historical and near-real-time data on freight rates, commodity prices, seasonal demand, and port infrastructure, and converts it into three outputs a logistics manager can act on directly: a rate forecast for a chosen route and vessel class, a ranked recommendation of the most suitable vessel type for a given cargo parcel, and an early-warning flag for volatility or congestion

risk. Rather than requiring a logistics manager to interpret raw market data, the platform packages the forecast, the constraint checks, and the risk signal into a single recommendation screen.

2.1 Solution Workflow

- **Data ingestion layer** — collects historical freight rate series (by vessel class and route), Baltic-type index proxies, commodity price indices for coal and other dry-bulk cargo, seasonal/trade-lane indicators, and a structured port infrastructure reference table (draft, LOA, beam, berth count, cargo-handling rate) for both origin ports and the seven target East Coast ports.
- **Forecasting engine** — time-series and regression models trained per route/vessel-class pair to project freight rates over the requested contract horizon, with confidence intervals rather than point estimates alone.
- **Constraint-matching engine** — cross-references the entered cargo volume and destination port against the infrastructure reference table to eliminate vessel classes that cannot physically call at the port, before ranking the remainder by forecast cost and turnaround efficiency.
- **Risk and idle-time module** — flags routes/periods with rising volatility or congestion signals, and surfaces alternative employment windows to reduce deadheading.
- **Dashboard interface** — lets a logistics manager enter cargo details, origin/destination ports, and desired contract duration, and returns the forecast, the recommended vessel type, and the risk flag on one screen.

3. Key Features

Feature	Description
Freight rate forecasting	Route- and vessel-class-specific rate projections over a selectable contract horizon (short-term / mid-term), with confidence bands.
Optimal market entry timing	Highlights the forecast window most favourable for fixing a charter, based on projected rate troughs relative to the requested horizon.
Vessel type recommendation	Ranks Handysize / Supramax / Panamax / Capesize options for a cargo parcel after filtering out vessels that violate port draft, LOA, beam, or handling-rate constraints.
Port constraint database	Structured reference data for East Coast Indian ports and key origin ports (Australia, US, Mozambique, Russia, Indonesia).
Idle-time / deadheading advisory	Forecasts low-demand periods and suggests repositioning or alternative employment to reduce unproductive vessel time.
Risk early-warning flags	Signals elevated volatility or congestion risk for a selected route before a contract is committed.

Feature	Description
Interactive dashboard	Web dashboard for cargo, route, and contract-duration input, returning a consolidated recommendation view.

4. Technical Approach and Methodology

4.1 Data Sources

- Historical freight rate series by vessel class and route (public shipping-index and brokerage-report data used as a proxy where live subscription feeds are not accessible during development).
- Commodity price trends for dry-bulk cargo (e.g., thermal coal) relevant to the chosen trade lanes.
- Seasonal and trade-lane demand indicators for Australia, the US, Mozambique, Russia, and Indonesia as origins.
- A curated port infrastructure dataset covering draft, LOA, beam, berth capacity, and cargo-handling rate for the seven target East Coast ports and the principal loading ports at each origin.

4.2 Modelling Approach

The forecasting engine will be built and compared across two families of models: classical time-series methods (ARIMA/SARIMA and Prophet) as an interpretable baseline, and gradient-boosted regression (LightGBM) with Optuna-based hyperparameter tuning for handling multiple exogenous features (commodity price, seasonality, port congestion proxies) together. Model selection will be based on rolling-window backtesting with MAPE/RMSE on held-out future periods, consistent with a forecasting problem rather than a single train/test split. Vessel-type recommendation is treated as a constrained ranking problem: a rule-based filter removes physically infeasible vessel classes for a given port, and the remaining options are ranked by forecast freight cost and expected turnaround time.

4.3 Proposed Technology Stack

Layer	Tools
Data processing	Python, Pandas, NumPy
Forecasting / ML	Scikit-learn, LightGBM, Prophet/statsmodels, Optuna for tuning
Backend / API	FastAPI
Dashboard	Streamlit
Storage	MySQL / PostgreSQL for rate history and the port-constraint reference table
Deployment	Docker-based packaging for portability across on-prem or cloud hosting

5. Target Users and Beneficiaries

- **Logistics and chartering managers** at bulk cargo importers/traders who fix vessels for East Coast Indian ports and currently rely on daily broker calls.
- **Ship-broking and chartering desks** seeking a quantitative second opinion alongside market intelligence when advising clients on contract timing.
- **Port operations and terminal planning teams**, who benefit indirectly from better-matched vessel calls and reduced congestion from mismatched vessel sizes.
- **Procurement and supply-chain planning teams** at power utilities, steel plants, and other coal/bulk-commodity end users that plan multi-voyage import schedules.

6. Practical Relevance and Scope for Future Implementation

The problem statement is drawn directly from an SIH logistics and transportation track and reflects an operational pain point rather than a constructed academic exercise: freight rate volatility and port-infrastructure mismatch are recurring, well-documented cost drivers in Indian dry-bulk imports. A working forecasting and recommendation layer has direct cost-saving relevance to any organisation chartering vessels into these ports, and the underlying approach generalises beyond coal to other bulk commodities and additional ports once the constraint database is extended.

6.1 Scope for Extension / Revenue Generation

- Packaging the platform as a subscription analytics tool for chartering desks and mid-size bulk importers who cannot justify an in-house quant team.
- Extending the port-constraint database and route coverage as a licensable dataset product independent of the forecasting model itself.
- Adding an API tier so the recommendation engine can be embedded directly into a client's existing procurement or ERP workflow.

7. Individual Contribution

This is an individual capstone submission; all components are undertaken by Ankit Ojha. The work builds directly on prior applied ML experience — including gradient-boosted forecasting with LightGBM and Optuna tuning on a Streamlit dashboard for a demand-forecasting use case, and FastAPI-backed regression deployment — extended here to a multi-route, multi-constraint freight forecasting context. The individual contribution spans the full pipeline:

- Problem scoping and translation of the SIH statement into a defined technical specification (Sections 1–4 of this proposal).
- Data collection and construction of the port-infrastructure constraint reference table for origin and East Coast destination ports.
- Design, training, and backtesting of the forecasting models (classical time-series baseline and LightGBM/Optuna regression).

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- Implementation of the constraint-matching and vessel-ranking logic.
 - Development of the FastAPI backend and Streamlit dashboard, and end-to-end integration and testing.

8. Indicative Timeline

Phase	Milestone
Phase 1	Data collection: freight rate history, commodity indices, port constraint dataset
Phase 2	Baseline time-series forecasting model and backtesting framework
Phase 3	LightGBM/Optuna forecasting model, constraint-matching and vessel-ranking logic
Phase 4	Risk/idle-time advisory module and FastAPI backend integration
Phase 5	Streamlit dashboard, end-to-end testing, and documentation

9. Conclusion

The proposed Intelligent Freight Forecasting Model directly addresses a genuine, well-scoped operational problem in bulk cargo procurement, moves beyond a basic prototype by combining forecasting, hard infrastructure constraints, and risk flagging into one decision-support tool, and has a clear path to practical deployment and extension beyond the academic submission.